

Table 1: Cross sections, branching ratios, uncertainties, and references.

Sample	Type	Value	Uncertainty	Reference	Comments
one_pb	σ [pb]	1	—	—	—
1pb	σ [pb]	1	—	—	—
BR_H_bb	BR	0.576	—	https://gitlab.cern.ch/hh/recommendations/-/blob/d7ad328c2ca55c75466b0a480af19823b7fa305b/CrossSections.md	—
BR_H_WW	BR	0.2203	—	https://gitlab.cern.ch/hh/recommendations/-/blob/d7ad328c2ca55c75466b0a480af19823b7fa305b/CrossSections.md	—
BR_H_ZZ	BR	0.02716	—	https://gitlab.cern.ch/hh/recommendations/-/blob/d7ad328c2ca55c75466b0a480af19823b7fa305b/CrossSections.md	—
BR_H_tautau	BR	0.06208	—	https://gitlab.cern.ch/hh/recommendations/-/blob/d7ad328c2ca55c75466b0a480af19823b7fa305b/CrossSections.md	—
BR_W_lnu	BR	0.1086	—	https://pdg.lbl.gov/2019/listings/rpp2019-list-w-boson.pdf	—
BR_W_qq	BR	0.8914 $= 1 - BR_W_l\nu$	—	https://pdg.lbl.gov/2019/listings/rpp2019-list-w-boson.pdf	—
BR_Z_ll	BR	0.033658	—	https://pdg.lbl.gov/2018/listings/rpp2018-list-z-boson.pdf	—
BR_Z_nunu	BR	0.2	—	https://pdg.lbl.gov/2018/listings/rpp2018-list-z-boson.pdf	—
DYto2E_MLL_10to50_powheg	σ [pb]	2094.2	total: $+0.7\%$ -1.1% (<i>relative</i>)	https://twiki.cern.ch/twiki/bin/viewauth/CMS/MATRIXCrossSectionsat13p6TeV	—
DYto2E_MLL_50to120_powheg	σ [pb]	6744	total: ± 1.132 (<i>absolute</i>)	XSDBDYto2E_MLL-10to50_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:18:40	—
DYto2E_MLL_120to200_powheg	σ [pb]	2219	total: ± 0.2327 (<i>absolute</i>)	XSDBDYto2E_MLL-50to120_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:19:01	—
DYto2E_MLL_20to200_powheg	σ [pb]	21.65	total: ± 0.003184 (<i>absolute</i>)	XSDBDYto2E_MLL-120to200_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:18:42	—
DYto2E_MLL_200to400_powheg	σ [pb]	3.058	total: ± 0.000465 (<i>absolute</i>)	XSDBDYto2E_MLL-200to400_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:18:48	—
DYto2E_MLL_400to800_powheg	σ [pb]	0.2691	total: $\pm 4.215e-05$ (<i>absolute</i>)	XSDBDYto2E_MLL-400to800_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:18:57	—
DYto2E_MLL_800to1500_powheg	σ [pb]	0.01915	total: $\pm 3.085e-06$ (<i>absolute</i>)	XSDBDYto2E_MLL-800to1500_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:19:05	—
DYto2E_MLL_1500to2500_powheg	σ [pb]	0.001111	total: $\pm 1.787e-07$ (<i>absolute</i>)	XSDBDYto2E_MLL-1500to2500_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:18:46	—
DYto2E_MLL_2500to4000_powheg	σ [pb]	$5.949e-05$	total: $\pm 9.162e-09$ (<i>absolute</i>)	XSDBDYto2E_MLL-2500to4000_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:18:53	—
DYto2E_MLL_4000to6000_powheg	σ [pb]	$1.558e-06$	total: $\pm 2.078e-10$ (<i>absolute</i>)	XSDBDYto2E_MLL-4000to6000_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:18:55	—
DYto2E_MLL_6000_powheg	σ [pb]	$3.519e-08$	total: $\pm 6.811e-12$ (<i>absolute</i>)	XSDBDYto2E_MLL-6000_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:19:03	—
DYto2Mu_MLL_50to130_powheg_minnlo	σ [pb]	2088	total: ± 16.55 (<i>absolute</i>)	XSDBDYto2Mu_Bin-MLL-50to130_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photosmodifiedOn2026-05-0614:44:54	—
DYto2Mu_MLL_130to200_powheg_minnlo	σ [pb]	13.12	total: ± 0.04125 (<i>absolute</i>)	XSDBDYto2Mu_Bin-MLL-130to200_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photosmodifiedOn2026-05-0614:44:54	—
DYto2Mu_MLL_200to400_powheg_minnlo	σ [pb]	3.058	total: ± 0.000465 (<i>absolute</i>)	XSDBDYto2Mu_MLL-200to400_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:11	—
DYto2Mu_MLL_400to600_powheg_minnlo	σ [pb]	0.2195	total: ± 0.0006713 (<i>absolute</i>)	XSDBDYto2Mu_Bin-MLL-400to600_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photosmodifiedOn2026-05-0614:46:01	—
DYto2Mu_MLL_600to800_powheg_minnlo	σ [pb]	0.04258	total: ± 0.0001269 (<i>absolute</i>)	XSDBDYto2Mu_Bin-MLL-600to800_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photosmodifiedOn2026-05-0614:46:01	—
DYto2Mu_MLL_800to1000_powheg_minnlo	σ [pb]	0.01222	total: $\pm 3.782e-05$ (<i>absolute</i>)	XSDBDYto2Mu_Bin-MLL-800to1000_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photosmodifiedOn2026-05-0614:46:01	—
DYto2Mu_MLL_1000to1500_powheg_minnlo	σ [pb]	0.006627	total: $\pm 2.065e-05$ (<i>absolute</i>)	XSDBDYto2Mu_Bin-MLL-1000to1500_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photosmodifiedOn2026-05-0614:44:56	—
DYto2Mu_MLL_1500to2000_powheg_minnlo	σ [pb]	0.0009183	total: $\pm 2.853e-06$ (<i>absolute</i>)	XSDBDYto2Mu_Bin-MLL-1500to2000_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photosmodifiedOn2026-05-0614:44:44	—
DYto2Mu_MLL_2000to4000_powheg_minnlo	σ [pb]	0.0002396	total: $\pm 7.424e-07$ (<i>absolute</i>)	XSDBDYto2Mu_Bin-MLL-2000to4000_TuneCP5_13p6TeV_powhegMINNLO-pythia8-photosmodifiedOn2026-05-0614:45:35	—
DYto2Mu_MLL_4000to6000_powheg_minnlo	σ [pb]	$1.558e-06$	total: $\pm 2.078e-10$ (<i>absolute</i>)	XSDBDYto2Mu_MLL-4000to6000_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:18	—

Table 1 – continued					
Sample	Type	Value	Uncertainty	Reference	Comments
DYto2Mu_MLL_6000to13600_powheg_minnl0	σ [pb]	3.519e − 08	total: $\pm 6.811e - 12$ (<i>absolute</i>)	XSDBDYto2Mu_MLL-6000_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:25	—
DYto2L_M_10to50_amcatnl0FXFX_allFlavors	σ [pb]	20950	total: ± 183.5 (<i>absolute</i>)	XSDBDYto2L-2Jets_MLL-10to50_TuneCP5_13p6TeV_amcatnl0FXFX-pythia8modifiedOn2023-03-2613:56:18	—
DYto2L_M_10to50_amcatnl0FXFX_singleFlavor	σ [pb]	6983.3333	total: ± 61.166667 (<i>absolute</i>)	XSDBDYto2L-2Jets_MLL-10to50_TuneCP5_13p6TeV_amcatnl0FXFX-pythia8modifiedOn2023-03-2613:56:18	—
DYto2L_M_10to50_madgraphMLM	σ [pb]	$\begin{matrix} = \\ DYto2L_M_10to50_amcatnl0FXFX_allFlavors/3 \end{matrix}$ 17380	total: ± 26.57 (<i>absolute</i>)	XSDBDYto2L-4Jets_MLL-10to50_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:19:38	—
DYto2L_M_50_0J_amcatnl0FXFX_allFlavors	σ [pb]	5378	total: ± 8.007 (<i>absolute</i>)	XSDBDYto2L-2Jets_MLL-50_0J_TuneCP5_13p6TeV_amcatnl0FXFX-pythia8modifiedOn2024-04-0319:19:07	—
DYto2L_M_50_0J_amcatnl0FXFX_singleFlavor	σ [pb]	1792.6667 $= 5378.0/3$	total: ± 2.669 (<i>absolute</i>)	XSDBDYto2L-2Jets_MLL-50_0J_TuneCP5_13p6TeV_amcatnl0FXFX-pythia8modifiedOn2024-04-0319:19:07	divided by 3 to account for e, mu, tau
DYto2Tau_M_50_0J_Filtered_amcatnl0FXFX_allFlavors	σ [pb]	847.035	total: ± 14.152 (<i>absolute</i>)	GenXSecAnalyzer/DYto2Tau-2Jets_M-50_0J_Filtered_TuneCP5_13p6TeV_amcatnl0FXFX-pythia8/Run3Summer22MiniA0Dv4-130X_mcRun3_2022_realistic_v5-v3/MINIAODSIM	—
DYto2Tau_M_50_0J_Filtered_amcatnl0FXFX_singleFlavor	σ [pb]	282.345 $= 847.035/3$	total: ± 4.7173333 (<i>absolute</i>)	GenXSecAnalyzer/DYto2Tau-2Jets_M-50_0J_Filtered_TuneCP5_13p6TeV_amcatnl0FXFX-pythia8/Run3Summer22MiniA0Dv4-130X_mcRun3_2022_realistic_v5-v3/MINIAODSIM	divided by 3 to account for e, mu, tau
DYto2L_M_50_1J_amcatnl0FXFX_allFlavors	σ [pb]	1017	total: ± 6.264 (<i>absolute</i>)	XSDBDYto2L-2Jets_MLL-50_1J_TuneCP5_13p6TeV_amcatnl0FXFX-pythia8modifiedOn2024-04-0319:19:09	—
DYto2L_M_50_1J_amcatnl0FXFX_singleFlavor	σ [pb]	339 $= 1017.0/3$	total: ± 2.088 (<i>absolute</i>)	XSDBDYto2L-2Jets_MLL-50_1J_TuneCP5_13p6TeV_amcatnl0FXFX-pythia8modifiedOn2024-04-0319:19:09	divided by 3 to account for e, mu, tau
DYto2Tau_M_50_1J_Filtered_amcatnl0FXFX_allFlavors	σ [pb]	210.112	total: ± 11.8 (<i>absolute</i>)	GenXSecAnalyzer/DYto2Tau-2Jets_M-50_1J_Filtered_TuneCP5_13p6TeV_amcatnl0FXFX-pythia8/Run3Summer22MiniA0Dv4-130X_mcRun3_2022_realistic_v5-v3/MINIAODSIM	—
DYto2Tau_M_50_1J_Filtered_amcatnl0FXFX_singleFlavor	σ [pb]	70.037333 $= 210.112/3$	total: ± 3.9333333 (<i>absolute</i>)	GenXSecAnalyzer/DYto2Tau-2Jets_M-50_1J_Filtered_TuneCP5_13p6TeV_amcatnl0FXFX-pythia8/Run3Summer22MiniA0Dv4-130X_mcRun3_2022_realistic_v5-v3/MINIAODSIM	divided by 3 to account for e, mu, tau
DYto2L_M_50_2J_amcatnl0FXFX_allFlavors	σ [pb]	385.5	total: ± 3.858 (<i>absolute</i>)	XSDBDYto2L-2Jets_MLL-50_2J_TuneCP5_13p6TeV_amcatnl0FXFX-pythia8modifiedOn2024-04-0319:19:12	—
DYto2L_M_50_2J_amcatnl0FXFX_singleFlavor	σ [pb]	128.5 $= 385.5/3$	total: ± 1.286 (<i>absolute</i>)	XSDBDYto2L-2Jets_MLL-50_2J_TuneCP5_13p6TeV_amcatnl0FXFX-pythia8modifiedOn2024-04-0319:19:12	divided by 3 to account for e, mu, tau
DYto2Tau_M_50_2J_Filtered_amcatnl0FXFX_allFlavors	σ [pb]	83.268	total: ± 8.875 (<i>absolute</i>)	GenXSecAnalyzer/DYto2Tau-2Jets_M-50_2J_Filtered_TuneCP5_13p6TeV_amcatnl0FXFX-pythia8/Run3Summer22MiniA0Dv4-130X_mcRun3_2022_realistic_v5-v4/MINIAODSIM	—
DYto2Tau_M_50_2J_Filtered_amcatnl0FXFX_singleFlavor	σ [pb]	27.756 $= 83.268/3$	total: ± 2.9583333 (<i>absolute</i>)	GenXSecAnalyzer/DYto2Tau-2Jets_M-50_2J_Filtered_TuneCP5_13p6TeV_amcatnl0FXFX-pythia8/Run3Summer22MiniA0Dv4-130X_mcRun3_2022_realistic_v5-v4/MINIAODSIM	divided by 3 to account for e, mu, tau
DYto2L_M_50_1J_madgraphMLM_allFlavors	σ [pb]	973.1	total: ± 2.613 (<i>absolute</i>)	XSDBDYto2L-4Jets_MLL-50_1J_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:19:40	—
DYto2L_M_50_1J_madgraphMLM_singleFlavor	σ [pb]	324.36667 $= 973.1/3$	total: ± 0.871 (<i>absolute</i>)	XSDBDYto2L-4Jets_MLL-50_1J_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:19:40	—
DYto2L_M_50_2J_madgraphMLM_allFlavors	σ [pb]	312.4	total: ± 0.915 (<i>absolute</i>)	XSDBDYto2L-4Jets_MLL-50_2J_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:19:42	—
DYto2L_M_50_2J_madgraphMLM_singleFlavor	σ [pb]	104.13333 $= 312.4/3$	total: ± 0.305 (<i>absolute</i>)	XSDBDYto2L-4Jets_MLL-50_2J_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:19:42	—
DYto2L_M_50_3J_madgraphMLM_allFlavors	σ [pb]	93.93	total: ± 0.2858 (<i>absolute</i>)	XSDBDYto2L-4Jets_MLL-50_3J_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:19:44	—
DYto2L_M_50_3J_madgraphMLM_singleFlavor	σ [pb]	31.31 $= 93.93/3$	total: ± 0.095266667 (<i>absolute</i>)	XSDBDYto2L-4Jets_MLL-50_3J_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:19:44	—
DYto2L_M_50_4J_madgraphMLM_allFlavors	σ [pb]	45.43	total: ± 0.1393 (<i>absolute</i>)	XSDBDYto2L-4Jets_MLL-50_4J_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:19:46	—
DYto2L_M_50_4J_madgraphMLM_singleFlavor	σ [pb]	15.143333 $= 45.43/3$	total: ± 0.046433333 (<i>absolute</i>)	XSDBDYto2L-4Jets_MLL-50_4J_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:19:46	—
DYto2L_M_50_amcatnl0FXFX_allFlavors	σ [pb]	6688	total: ± 83.99 (<i>absolute</i>)	XSDBDYto2L-2Jets_MLL-50_TuneCP5_13p6TeV_amcatnl0FXFX-pythia8modifiedOn2023-03-2613:56:19	—
DYto2L_M_50_amcatnl0FXFX_singleFlavor	σ [pb]	2229.3333 $= 6688.0/3$	total: ± 27.996667 (<i>absolute</i>)	XSDBDYto2L-2Jets_MLL-50_TuneCP5_13p6TeV_amcatnl0FXFX-pythia8modifiedOn2023-03-2613:56:19	divided by 3 to account for e, mu, tau
DYto2L_M_50_madgraphMLM_allFlavors	σ [pb]	5467	total: ± 13.22 (<i>absolute</i>)	XSDBDYto2L-4Jets_MLL-50_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:19:49	—
DYto2L_M_50_madgraphMLM_singleFlavor	σ [pb]	1822.3333 $= 5467.0/3$	total: ± 4.4066667 (<i>absolute</i>)	XSDBDYto2L-4Jets_MLL-50_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:19:49	—

Table 1 – continued					
Sample	Type	Value	Uncertainty	Reference	Comments
DYto2L_M_50_PTLL_40to100_1J_amcatnloFXFX_allFlavors	σ [pb]	475.3	total: ± 2.811 (<i>absolute</i>)	XSDBDYto2L-2Jets_MLL-50_PTLL-40to100_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:27	—
DYto2L_M_50_PTLL_40to100_1J_amcatnloFXFX_singleFlavor	σ [pb]	158.43333 = 475.3/3	total: ± 0.937 (<i>absolute</i>)	XSDBDYto2L-2Jets_MLL-50_PTLL-40to100_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:27	divided by 3 to account for e, mu, tau
DYto2L_M_50_PTLL_40to100_2J_amcatnloFXFX_allFlavors	σ [pb]	179.3	total: ± 1.827 (<i>absolute</i>)	XSDBDYto2L-2Jets_MLL-50_PTLL-40to100_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:29	—
DYto2L_M_50_PTLL_40to100_2J_amcatnloFXFX_singleFlavor	σ [pb]	59.766667 = 179.3/3	total: ± 0.609 (<i>absolute</i>)	XSDBDYto2L-2Jets_MLL-50_PTLL-40to100_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:29	divided by 3 to account for e, mu, tau
DYto2L_M_50_PTLL_100to200_1J_amcatnloFXFX_allFlavors	σ [pb]	45.42	total: ± 0.2314 (<i>absolute</i>)	XSDBDYto2L-2Jets_MLL-50_PTLL-100to200_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:15	—
DYto2L_M_50_PTLL_100to200_1J_amcatnloFXFX_singleFlavor	σ [pb]	15.14 = 45.42/3	total: ± 0.077133333 (<i>absolute</i>)	XSDBDYto2L-2Jets_MLL-50_PTLL-100to200_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:15	divided by 3 to account for e, mu, tau
DYto2L_M_50_PTLL_100to200_2J_amcatnloFXFX_allFlavors	σ [pb]	51.68	total: ± 0.4864 (<i>absolute</i>)	XSDBDYto2L-2Jets_MLL-50_PTLL-100to200_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:16	—
DYto2L_M_50_PTLL_100to200_2J_amcatnloFXFX_singleFlavor	σ [pb]	17.226667 = 51.68/3	total: ± 0.16213333 (<i>absolute</i>)	XSDBDYto2L-2Jets_MLL-50_PTLL-100to200_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:16	divided by 3 to account for e, mu, tau
DYto2L_M_50_PTLL_200to400_1J_amcatnloFXFX_allFlavors	σ [pb]	3.382	total: ± 0.01443 (<i>absolute</i>)	XSDBDYto2L-2Jets_MLL-50_PTLL-200to400_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:18	—
DYto2L_M_50_PTLL_200to400_1J_amcatnloFXFX_singleFlavor	σ [pb]	1.1273333 = 3.382/3	total: ± 0.00481 (<i>absolute</i>)	XSDBDYto2L-2Jets_MLL-50_PTLL-200to400_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:18	divided by 3 to account for e, mu, tau
DYto2L_M_50_PTLL_200to400_2J_amcatnloFXFX_allFlavors	σ [pb]	7.159	total: ± 0.05581 (<i>absolute</i>)	XSDBDYto2L-2Jets_MLL-50_PTLL-200to400_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:19	—
DYto2L_M_50_PTLL_200to400_2J_amcatnloFXFX_singleFlavor	σ [pb]	2.3863333 = 7.159/3	total: ± 0.018603333 (<i>absolute</i>)	XSDBDYto2L-2Jets_MLL-50_PTLL-200to400_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:19	divided by 3 to account for e, mu, tau
DYto2L_M_50_PTLL_400to600_1J_amcatnloFXFX_allFlavors	σ [pb]	0.1162	total: ± 0.0004367 (<i>absolute</i>)	XSDBDYto2L-2Jets_MLL-50_PTLL-400to600_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:21	—
DYto2L_M_50_PTLL_400to600_1J_amcatnloFXFX_singleFlavor	σ [pb]	0.038733333 = 0.1162/3	total: ± 0.00014556667 (<i>absolute</i>)	XSDBDYto2L-2Jets_MLL-50_PTLL-400to600_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:21	divided by 3 to account for e, mu, tau
DYto2L_M_50_PTLL_400to600_2J_amcatnloFXFX_allFlavors	σ [pb]	0.4157	total: ± 0.002364 (<i>absolute</i>)	XSDBDYto2L-2Jets_MLL-50_PTLL-400to600_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:23	—
DYto2L_M_50_PTLL_400to600_2J_amcatnloFXFX_singleFlavor	σ [pb]	0.13856667 = 0.4157/3	total: ± 0.000788 (<i>absolute</i>)	XSDBDYto2L-2Jets_MLL-50_PTLL-400to600_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:23	divided by 3 to account for e, mu, tau
DYto2L_M_50_PTLL_600_1J_amcatnloFXFX_allFlavors	σ [pb]	0.01392	total: $\pm 5.797e-05$ (<i>absolute</i>)	XSDBDYto2L-2Jets_MLL-50_PTLL-600_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:34	—
DYto2L_M_50_PTLL_600_1J_amcatnloFXFX_singleFlavor	σ [pb]	0.00464 = 0.01392/3	total: $5.797e-05 / 3$ (<i>absolute</i>)	XSDBDYto2L-2Jets_MLL-50_PTLL-600_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:34	divided by 3 to account for e, mu, tau
DYto2L_M_50_PTLL_600_2J_amcatnloFXFX_allFlavors	σ [pb]	0.07019	total: ± 0.0003334 (<i>absolute</i>)	XSDBDYto2L-2Jets_MLL-50_PTLL-600_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:36	—
DYto2L_M_50_PTLL_600_2J_amcatnloFXFX_singleFlavor	σ [pb]	0.023396667 = 0.07019/3	total: ± 0.00011113333 (<i>absolute</i>)	XSDBDYto2L-2Jets_MLL-50_PTLL-600_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0319:19:36	divided by 3 to account for e, mu, tau
DYto2Mu_MLL_105to160_amcatnloFXFX	σ [pb]	49.03	total: ± 0.209 (<i>absolute</i>)	XSDBDYto2Mu-2Jets_MLL-105To160_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-12-2007:59:27	—
DYto2Mu_MLL_105to160_amcatnloFXFX_VBFFilter	σ [pb]	3.105	total: ± 0.006282 (<i>absolute</i>)	XSDBDYto2Mu-2Jets_Bin-MLL-105to160_Fil-VBF_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2026-05-0614:44:56	—
DYto2MuMu-M105-180_13p6TeV_MiNNLO_pythia8	σ [pb]	48.6	total: ± 0.01 (<i>absolute</i>)	internalfromprivateprod	—
DYto2Mu_MLL_10to50_powheg	σ [pb]	6744	total: ± 1.132 (<i>absolute</i>)	XSDBDYto2Mu_MLL-10to50_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:02	—
DYto2Mu_MLL_50to120_powheg	σ [pb]	2219	total: ± 0.2327 (<i>absolute</i>)	XSDBDYto2Mu_MLL-50to120_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:24	—
DYto2Mu_MLL_120to200_powheg	σ [pb]	21.65	total: ± 0.003184 (<i>absolute</i>)	XSDBDYto2Mu_MLL-120to200_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:04	—

Continued on next page

Table 1 – continued					
Sample	Type	Value	Uncertainty	Reference	Comments
DYto2Mu_MLL_200to400_powheg	σ [pb]	3.058	total: ± 0.000465 (<i>absolute</i>)	XSDBDYto2Mu_MLL-200to400_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:11	—
DYto2Mu_MLL_400to800_powheg	σ [pb]	0.2691	total: $\pm 4.215e - 05$ (<i>absolute</i>)	XSDBDYto2Mu_MLL-400to800_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:21	—
DYto2Mu_MLL_800to1500_powheg	σ [pb]	0.01915	total: $\pm 3.085e - 06$ (<i>absolute</i>)	XSDBDYto2Mu_MLL-800to1500_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:27	—
DYto2Mu_MLL_1500to2500_powheg	σ [pb]	0.001111	total: $\pm 1.787e - 07$ (<i>absolute</i>)	XSDBDYto2Mu_MLL-1500to2500_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:10	—
DYto2Mu_MLL_2500to4000_powheg	σ [pb]	$5.949e - 05$	total: $\pm 9.162e - 09$ (<i>absolute</i>)	XSDBDYto2Mu_MLL-2500to4000_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:16	—
DYto2Mu_MLL_4000to6000_powheg	σ [pb]	$1.558e - 06$	total: $\pm 2.078e - 10$ (<i>absolute</i>)	XSDBDYto2Mu_MLL-4000to6000_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:18	—
DYto2Mu_MLL_6000_powheg	σ [pb]	$3.519e - 08$	total: $\pm 6.811e - 12$ (<i>absolute</i>)	XSDBDYto2Mu_MLL-6000_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:25	—
DYto2Tau_MLL_10to50_powheg	σ [pb]	6744	total: ± 1.132 (<i>absolute</i>)	XSDBDYto2Tau_MLL-10to50_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:29	—
DYto2Tau_MLL_50to120_powheg	σ [pb]	2219	total: ± 0.2327 (<i>absolute</i>)	XSDBDYto2Tau_MLL-50to120_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:45	—
DYto2Tau_MLL_120to200_powheg	σ [pb]	21.65	total: ± 0.003184 (<i>absolute</i>)	XSDBDYto2Tau_MLL-120to200_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:31	—
DYto2Tau_MLL_200to400_powheg	σ [pb]	3.058	total: ± 0.000465 (<i>absolute</i>)	XSDBDYto2Tau_MLL-200to400_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:36	—
DYto2Tau_MLL_400to800_powheg	σ [pb]	0.2691	total: $\pm 4.215e - 05$ (<i>absolute</i>)	XSDBDYto2Tau_MLL-400to800_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:43	—
DYto2Tau_MLL_800to1500_powheg	σ [pb]	0.01915	total: $\pm 3.085e - 06$ (<i>absolute</i>)	XSDBDYto2Tau_MLL-800to1500_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:49	—
DYto2Tau_MLL_1500to2500_powheg	σ [pb]	0.001111	total: $\pm 1.787e - 07$ (<i>absolute</i>)	XSDBDYto2Tau_MLL-1500to2500_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:34	—
DYto2Tau_MLL_2500to4000_powheg	σ [pb]	$5.949e - 05$	total: $\pm 9.162e - 09$ (<i>absolute</i>)	XSDBDYto2Tau_MLL-2500to4000_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:38	—
DYto2Tau_MLL_4000to6000_powheg	σ [pb]	$1.558e - 06$	total: $\pm 2.078e - 10$ (<i>absolute</i>)	XSDBDYto2Tau_MLL-4000to6000_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:41	—
DYto2Tau_MLL_6000_powheg	σ [pb]	$3.519e - 08$	total: $\pm 6.811e - 12$ (<i>absolute</i>)	XSDBDYto2Tau_MLL-6000_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0319:20:47	—
DYto2TautoMuTauh_M_50_madgraphMLM	σ [pb]	53.28	total: ± 0.5212 (<i>absolute</i>)	XSDBDYto2TautoMuTauh_M-50_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2023-03-2613:56:19	—
LLJJ_SM_M105-160_MJJ120_13p6TeV_dipole	σ [pb]	0.06	—	internalfromprivateprod	—
LLJJ_SM_M105-160_MJJ120_13p6TeV_dipole_vbffilter	σ [pb]	0.06	—	internalfromprivateprod	—
LLJJ_SM_M-50_MJJ120_13p6TeV_dipole	σ [pb]	0.88	—	internalfromprivateprod	—
LLJJ_SM_M-50_MJJ120_13p6TeV_herwig7	σ [pb]	0.88	—	internalfromprivateprod	—
GluGluHto2Mu_M120	σ [pb]	0.013595453 $= 56.11 \times (0.0002423)$	theory: $+4.6\%$ (<i>relative</i>) -6.9% (<i>relative</i>) theory (Gaussian): $\pm 4\%$ (<i>relative</i>) PDF + α_s: $\pm 3.2\%$ (<i>relative</i>) pdf: $\pm 1.9\%$ (<i>relative</i>) α_s : $\pm 2.6\%$ (<i>relative</i>) theory (BR): $+1.23\%$ 1.23% (<i>relative</i>) m_q (BR): $+1.07\%$ (<i>relative</i>) -1.09% (<i>relative</i>) α_s (BR): $+0.69\%$ (<i>relative</i>) -0.70% (<i>relative</i>)	https://arxiv.org/pdf/2402.09955forXSandhttps://twiki.cern.ch/twiki/bin/view/LHCPhysics/CERNYellowReportPageBRforBR	—
GluGluHto2Mu	σ [pb]	0.011365248 $= 52.23 \times (0.0002176)$	theory: $+4.6\%$ (<i>relative</i>) -6.7% (<i>relative</i>) theory (Gaussian): $\pm - 3.9$ (<i>absolute</i>) PDF + α_s: $\pm - 3.2$ (<i>absolute</i>) pdf: $\pm - 1.9$ (<i>absolute</i>) α_s : $\pm - 2.6$ (<i>absolute</i>) theory (BR): $+1.23\%$ (<i>relative</i>) -1.2% (<i>relative</i>) m_q (BR): $+0.97\%$ (<i>relative</i>) -0.99% (<i>relative</i>) α_s (BR): $+0.59\%$ (<i>relative</i>) -0.64% (<i>relative</i>)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LCHHWG136TeVxsec_extrapandhttps://twiki.cern.ch/twiki/bin/view/LHCPhysics/CERNYellowReportPageBRforBR	—

Table 1 – continued					
Sample	Type	Value	Uncertainty	Reference	Comments
GluGluHto2Mu_M125	σ [pb]	0.011365248 = $52.23 \times (0.0002176)$	theory: $+4.6\%$ <i>(relative)</i> -6.7% theory (Gaussian): ± 3.9 <i>(absolute)</i> PDF + α_s : ± 3.2 <i>(absolute)</i> pdf: ± 1.9 <i>(absolute)</i> α_s : ± 2.6 <i>(absolute)</i> theory (BR): $+1.23\%$ <i>(relative)</i> -1.2% m_q (BR): $+0.97\%$ <i>(relative)</i> -0.99% α_s (BR): $+0.59\%$ <i>(relative)</i> -0.64%	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LCHWVG136TeVxsec_extrapandhttps://twiki.cern.ch/twiki/bin/view/LHCPhysics/CERNYellowReportPageBRforBR	—
GluGluHto2Mu_M130	σ [pb]	0.009150375 = $48.75 \times (0.0001877)$	theory: $+4.5\%$ <i>(relative)</i> -6.5% theory (Gaussian): $\pm 3.8\%$ <i>(relative)</i> PDF + α_s : $\pm 3.2\%$ <i>(relative)</i> pdf: $\pm 1.8\%$ <i>(relative)</i> α_s : $\pm 2.6\%$ <i>(relative)</i> theory (BR): $+1.22\%$ 1.21% <i>(relative)</i> m_q (BR): $+0.86\%$ <i>(relative)</i> -0.82% α_s (BR): $+0.52\%$ <i>(relative)</i> -0.47%	https://arxiv.org/pdf/2402.09955forXSandhttps://twiki.cern.ch/twiki/bin/view/LHCPhysics/CERNYellowReportPageBRforBR	—
GluGluHToTauTau_M125	σ [pb]	3.2758656 = $52.23 \times (0.06272)$	theory: $+4.6\%$ <i>(relative)</i> -6.7% theory (Gaussian): ± 3.9 <i>(absolute)</i> PDF + α_s : ± 3.2 <i>(absolute)</i> pdf: ± 1.9 <i>(absolute)</i> α_s : ± 2.6 <i>(absolute)</i>	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LCHWVG136TeVxsec_extrap	—
GluGluHToTauTau_M125_Filtered	σ [pb]	1.2762772 = $52.23 \times (0.06272) \times 3.896e-01$	theory: $+4.6\%$ <i>(relative)</i> -6.7% theory (Gaussian): ± 3.9 <i>(absolute)</i> PDF + α_s : ± 3.2 <i>(absolute)</i> pdf: ± 1.9 <i>(absolute)</i> α_s : ± 2.6 <i>(absolute)</i> filter efficiency: ± 0.534 <i>(absolute)</i>	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LCHWVG136TeVxsec_extrap*filterefffromgenXsec	—
GluGluHToWWTo2L2Nu_M125	σ [pb]	3.6364333 = $52.23 \times (0.2137) \times (3 \times 0.1086)$	theory: $+4.6\%$ <i>(relative)</i> -6.7% theory (Gaussian): ± 3.9 <i>(absolute)</i> PDF + α_s : ± 3.2 <i>(absolute)</i> pdf: ± 1.9 <i>(absolute)</i> α_s : ± 2.6 <i>(absolute)</i>	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LCHWVG136TeVxsec_extrap	—
GluGluHToBB_M125	σ [pb]	30.418752 = $52.23 \times (0.5824)$	theory: $+4.6\%$ <i>(relative)</i> -6.7% theory (Gaussian): ± 3.9 <i>(absolute)</i> PDF + α_s : ± 3.2 <i>(absolute)</i> pdf: ± 1.9 <i>(absolute)</i> α_s : ± 2.6 <i>(absolute)</i>	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LCHWVG136TeVxsec_extrapandhttps://twiki.cern.ch/twiki/bin/view/LHCPhysics/CERNYellowReportPageBRforBR	—
GluGlutoContinto2Zto2E2Nu	σ [pb]	23.31	total: $\pm 8.361e-05$ <i>(absolute)</i>	genXSecanalyzer	—
GluGlutoContinto2Zto2E2Mu	σ [pb]	6.115	total: $\pm 7.528e-05$ <i>(absolute)</i>	genXSecanalyzer	—
GluGlutoContinto2Zto2E2Tau	σ [pb]	6.242	total: ± 0.0006294 <i>(absolute)</i>	genXSecanalyzer	—
GluGlutoContinto2Zto2Mu2Nu	σ [pb]	23.31	total: $\pm 7.103e-05$ <i>(absolute)</i>	genXSecanalyzer	—
GluGlutoContinto2Zto2Mu2Tau	σ [pb]	6.242	total: ± 0.0006294 <i>(absolute)</i>	genXSecanalyzer	—
GluGlutoContinto2Zto4E	σ [pb]	3.06	total: ± 0.004207 <i>(absolute)</i>	genXSecanalyzer	—
GluGlutoContinto2Zto4Mu	σ [pb]	3.042	total: ± 0.003596 <i>(absolute)</i>	genXSecanalyzer	—
GluGlutoContinto2Zto4Tau	σ [pb]	3.042	total: ± 0.003596 <i>(absolute)</i>	genXSecanalyzer	—
VBFHToBB_M125	σ [pb]	2.3750272 = $4.078 \times (0.5824)$	QCD scale: $+0.5\%$ <i>(relative)</i> -0.3% PDF + α_s : $\pm 2.1\%$ <i>(relative)</i> pdf: $\pm 2.1\%$ <i>(relative)</i> α_s : ± 0.5 <i>(absolute)</i>	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LCHWVG136TeVxsec_extrapandhttps://twiki.cern.ch/twiki/bin/view/LHCPhysics/CERNYellowReportPageBRforBR	—
VBFHto2Tau	σ [pb]	4.18	total: ± 0.006886 <i>(absolute)</i>	XSDBVBFHto2TauUncorrelatedDecay_M-125_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2025-09-0412:43:24	—
VBFHto2Tau_Filtered	σ [pb]	1.717	total: ± 0.005038 <i>(absolute)</i>	XSDBVBFHto2TauUncorrelatedDecay_Filtered_M-125_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2025-09-0412:43:22	—
WminusH_Hbb_Wlnu	σ [pb]	0.10771876 = $0.5677 \times 0.5824 \times 3 \times 0.1086$	QCD scale: $+0.4\%$ <i>(relative)</i> -0.7% PDF + α_s : $\pm 1.8\%$ <i>(relative)</i> pdf: ± 1.6 <i>(absolute)</i> α_s : ± 0.9 <i>(absolute)</i>	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LCHWVG136TeVxsec_extrapandhttps://twiki.cern.ch/twiki/bin/view/LHCPhysics/CERNYellowReportPageBRforBR	—
WminusH_Hbb_Wqq	σ [pb]	0.22290972 = $0.5677 \times 0.5824 \times (1 - 3 \times 0.1086)$	QCD scale: $+0.4\%$ <i>(relative)</i> -0.7% PDF + α_s : $\pm 1.8\%$ <i>(relative)</i> pdf: ± 1.6 <i>(absolute)</i> α_s : ± 0.9 <i>(absolute)</i>	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LCHWVG136TeVxsec_extrapandhttps://twiki.cern.ch/twiki/bin/view/LHCPhysics/CERNYellowReportPageBRforBR	—

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Table 1 – continued

Sample	Type	Value	Uncertainty	Reference	Comments
WminusHto2Tau_Filtered	σ [pb]	0.2307	total: ± 0.0005923 (<i>absolute</i>)	XSDBWminusHto2TauUncorrelatedDecay_Filtered_M-125_TuneCP5_13p6TeV_powheg-minnlo-pythia8modifiedOn2025-09-0412:43:26	—
WminusHto2Tau	σ [pb]	0.5788	total: $\pm 1.752e - 05$ (<i>absolute</i>)	XSDBWminusHto2TauUncorrelatedDecay_M-125_TuneCP5_13p6TeV_powheg-minnlo-pythia8modifiedOn2025-09-0412:43:28	—
WplusH_Hbb_Wlnu	σ [pb]	0.16866515 $= 0.8889 \times 0.5824 \times 3 \times 0.1086$	QCD scale: $+0.4\%$ -0.7% (<i>relative</i>) PDF $+ \alpha_s$: $\pm 1.8\%$ (<i>relative</i>) pdf: ± 1.6 (<i>absolute</i>) α_s : ± 0.9 (<i>absolute</i>)	https: //indico.cern.ch/event/1119741/contributions/4715908/attachments/2383849/4073592/YR4_13p6_VH_update.pdf	—
WplusH_Hbb_Wqq	σ [pb]	0.34903021 $= 0.8889 \times 0.5824 \times (1 - 3 \times 0.1086)$	QCD scale: $+0.4\%$ -0.7% (<i>relative</i>) PDF $+ \alpha_s$: $\pm 1.8\%$ (<i>relative</i>) pdf: ± 1.6 (<i>absolute</i>) α_s : ± 0.9 (<i>absolute</i>)	https: //indico.cern.ch/event/1119741/contributions/4715908/attachments/2383849/4073592/YR4_13p6_VH_update.pdf	—
WplusHto2Tau_Filtered	σ [pb]	0.3504	total: ± 0.0009183 (<i>absolute</i>)	XSDBWplusHto2TauUncorrelatedDecay_Filtered_M-125_TuneCP5_13p6TeV_powheg-minnlo-pythia8modifiedOn2025-09-0412:43:29	—
WplusHto2Tau	σ [pb]	0.9273	total: ± 0.0001175 (<i>absolute</i>)	XSDBWplusHto2TauUncorrelatedDecay_M-125_TuneCP5_13p6TeV_powheg-minnlo-pythia8modifiedOn2025-09-0412:43:31	—
QCD_HT40to70_MLM	σ [pb]	$3.114e + 08$	total: ± 843600 (<i>absolute</i>)	XSDBQCD-4Jets_HT-40to70_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:46:52	—
QCD_HT70to100_MLM	σ [pb]	58500000	total: ± 167400 (<i>absolute</i>)	XSDBQCD-4Jets_HT-70to100_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:46:55	—
QCD_HT100to200_MLM	σ [pb]	25060000	total: ± 79510 (<i>absolute</i>)	GenXSecAnalyzer"root://cms-xrd-global.cern.ch//store/mc/Run3Summer22EEMiniAODv4/QCD-4Jets_HT-100to200_TuneCP5_13p6TeV_madgraphMLM-pythia8/MINIAODSIM/130X_mcRun3_2022_realistic_postEE_v6-v2/2520000/00527881-de92-4352-a197-c253c31c6aa7.root"	—
QCD_HT200to400_MLM	σ [pb]	1961000	total: ± 5862 (<i>absolute</i>)	XSDBQCD-4Jets_HT-200to400_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:46:48	—
QCD_HT400to600_MLM	σ [pb]	95620	total: ± 289.3 (<i>absolute</i>)	XSDBQCD-4Jets_HT-400to600_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:46:50	—
QCD_HT600to800_MLM	σ [pb]	13540	total: ± 41.11 (<i>absolute</i>)	XSDBQCD-4Jets_HT-600to800_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:46:54	—
QCD_HT800to1000_MLM	σ [pb]	3033	total: ± 9.206 (<i>absolute</i>)	XSDBQCD-4Jets_HT-800to1000_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:46:57	—
QCD_HT1000to1200_MLM	σ [pb]	883.7	total: ± 2.69 (<i>absolute</i>)	XSDBQCD-4Jets_HT-1000to1200_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:46:40	—
QCD_HT1200to1500_MLM	σ [pb]	383.5	total: ± 1.168 (<i>absolute</i>)	XSDBQCD-4Jets_HT-1200to1500_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:46:42	—
QCD_HT1500to2000_MLM	σ [pb]	125.2	total: ± 0.381 (<i>absolute</i>)	XSDBQCD-4Jets_HT-1500to2000_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:46:43	—
QCD_HT2000toInf_MLM	σ [pb]	26.49	total: ± 0.0807 (<i>absolute</i>)	XSDBQCD-4Jets_HT-2000_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0319:46:45	—
QCD_PT_10to30_EMEnriched	σ [pb]	6854000	total: ± 68270 (<i>absolute</i>)	XSDBQCD_PT-10to30_EMEnriched_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:28	—
QCD_PT_30to50_EMEnriched	σ [pb]	6698000	total: ± 62870 (<i>absolute</i>)	XSDBQCD_PT-30to50_EMEnriched_TuneCP5_13p6TeV_pythia8modifiedOn2023-05-2621:19:12	—
QCD_PT_50to80_EMEnriched	σ [pb]	2113000	total: ± 19470 (<i>absolute</i>)	XSDBQCD_PT-50to80_EMEnriched_TuneCP5_13p6TeV_pythia8modifiedOn2023-05-2621:19:12	—
QCD_PT_80to120_EMEnriched	σ [pb]	394100	total: ± 3654 (<i>absolute</i>)	XSDBQCD_PT-80to120_EMEnriched_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:48	—
QCD_PT_120to170_EMEnriched	σ [pb]	71600	total: ± 330.6 (<i>absolute</i>)	GenXSecAnalyzer"root://cms-xrd-global.cern.ch//store/mc/Run3Summer22EEMiniAODv4/QCD_PT-120to170_EMEnriched_TuneCP5_13p6TeV_pythia8/MINIAODSIM/130X_mcRun3_2022_realistic_postEE_v6-v2/2520000/4ef4de22-1e29-41db-acf0-9699123caa6d.root"	—
QCD_PT_170to300_EMEnriched	σ [pb]	17920	total: ± 164.5 (<i>absolute</i>)	XSDBQCD_PT-170to300_EMEnriched_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:33	—
QCD_PT_300_EMEnriched	σ [pb]	1231	total: ± 11.28 (<i>absolute</i>)	XSDBQCD_PT-300_EMEnriched_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:37	—
QCD_PT_300toInf_EMEnriched	σ [pb]	1224	total: ± 11.46 (<i>absolute</i>)	XSDBQCD_PT-300toInf_EMEnriched_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:39	—
QCD_PT_15to20_MuEnrichedPt5	σ [pb]	2960000	total: ± 29320 (<i>absolute</i>)	XSDBQCD_PT-15to20_MuEnrichedPt5_TuneCP5_13p6TeV_pythia8modifiedOn2023-05-2621:19:11	—
QCD_PT_20to30_MuEnrichedPt5	σ [pb]	2679000	total: ± 26580 (<i>absolute</i>)	XSDBQCD_PT-20to30_MuEnrichedPt5_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:54	—

Table 1 – continued					
Sample	Type	Value	Uncertainty	Reference	Comments
QCD_PT_30to50_MuEnrichedPt5	σ [pb]	1465000	total: ± 14360 (<i>absolute</i>)	XSDBQCD_PT-30to50_MuEnrichedPt5_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:41	—
QCD_PT_50to80_MuEnrichedPt5	σ [pb]	409500	total: ± 4049 (<i>absolute</i>)	XSDBQCD_PT-50to80_MuEnrichedPt5_TuneCP5_13p6TeV_pythia8modifiedOn2023-05-2621:19:13	—
QCD_PT_80to120_MuEnrichedPt5	σ [pb]	96200	total: ± 916.9 (<i>absolute</i>)	XSDBQCD_PT-80to120_MuEnrichedPt5_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:49	—
QCD_PT_120to170_MuEnrichedPt5	σ [pb]	22980	total: ± 215.1 (<i>absolute</i>)	XSDBQCD_PT-120to170_MuEnrichedPt5_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:51	—
QCD_PT_170to300_MuEnrichedPt5	σ [pb]	7763	total: ± 23.67 (<i>absolute</i>)	XSDBQCD_PT-170to300_MuEnrichedPt5_TuneCP5_13p6TeV_pythia8modifiedOn2024-04-0319:47:43	—
QCD_PT_300to470_MuEnrichedPt5	σ [pb]	699.1	total: ± 6.639 (<i>absolute</i>)	XSDBQCD_PT-300to470_MuEnrichedPt5_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:38	—
QCD_PT_470to600_MuEnrichedPt5	σ [pb]	68.24	total: ± 0.2049 (<i>absolute</i>)	XSDBQCD_PT-470to600_MuEnrichedPt5_TuneCP5_13p6TeV_pythia8modifiedOn2024-04-0319:47:45	—
QCD_PT_600to800_MuEnrichedPt5	σ [pb]	21.37	total: ± 0.199 (<i>absolute</i>)	XSDBQCD_PT-600to800_MuEnrichedPt5_TuneCP5_13p6TeV_pythia8modifiedOn2023-05-2621:19:13	—
QCD_PT_800to1000_MuEnrichedPt5	σ [pb]	3.913	total: ± 0.03526 (<i>absolute</i>)	XSDBQCD_PT-800to1000_MuEnrichedPt5_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:47	—
QCD_PT_1000_MuEnrichedPt5	σ [pb]	1.323	total: ± 0.003921 (<i>absolute</i>)	XSDBQCD_PT-1000_MuEnrichedPt5_TuneCP5_13p6TeV_pythia8modifiedOn2024-04-0319:47:38	—
QCD_PT_15to30	σ [pb]	$1.301e + 09$	total: ± 2135000 (<i>absolute</i>)	XSDBQCD_PT-15to30_TuneCP5_13p6TeV_pythia8modifiedOn2024-04-0320:23:32	—
QCD_PT_30to50	σ [pb]	$1.133e + 08$	total: ± 589300 (<i>absolute</i>)	XSDBQCD_PT-30to50_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:27	—
QCD_PT_50to80	σ [pb]	16760000	total: ± 27450 (<i>absolute</i>)	XSDBQCD_PT-50to80_TuneCP5_13p6TeV_pythia8modifiedOn2024-04-0319:47:46	—
QCD_PT_80to120	σ [pb]	2534000	total: ± 13280 (<i>absolute</i>)	XSDBQCD_PT-80to120_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:49	—
QCD_PT_120to170	σ [pb]	445800	total: ± 2174 (<i>absolute</i>)	XSDBQCD_PT-120to170_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:29	—
QCD_PT_170to300	σ [pb]	113700	total: ± 547.6 (<i>absolute</i>)	XSDBQCD_PT-170to300_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:34	—
QCD_PT_300to470	σ [pb]	7559	total: ± 36.67 (<i>absolute</i>)	XSDBQCD_PT-300to470_TuneCP5_13p6TeV_pythia8modifiedOn2023-05-2621:19:12	—
QCD_PT_470to600	σ [pb]	626.4	total: ± 2.863 (<i>absolute</i>)	XSDBQCD_PT-470to600_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:44	—
QCD_PT_600to800	σ [pb]	178.6	total: ± 0.8268 (<i>absolute</i>)	XSDBQCD_PT-600to800_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:45	—
QCD_PT_800to1000	σ [pb]	30.57	total: ± 0.1405 (<i>absolute</i>)	XSDBQCD_PT-800to1000_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:47	—
QCD_PT_1000to1400	σ [pb]	8.92	total: ± 0.04204 (<i>absolute</i>)	XSDBQCD_PT-1000to1400_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:26	—
QCD_PT_1400to1800	σ [pb]	0.8103	total: ± 0.003798 (<i>absolute</i>)	XSDBQCD_PT-1400to1800_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:30	—
QCD_PT_1800to2400	σ [pb]	0.1148	total: ± 0.0005556 (<i>absolute</i>)	XSDBQCD_PT-1800to2400_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:35	—
QCD_PT_2400to3200	σ [pb]	0.007542	total: $\pm 3.779e - 05$ (<i>absolute</i>)	XSDBQCD_PT-2400to3200_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:36	—
QCD_PT_3200	σ [pb]	0.0002331	total: $\pm 1.208e - 06$ (<i>absolute</i>)	XSDBQCD_PT-3200_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2613:58:43	—
TbarBtoLminusNuB_s_channel_4FS	σ [pb]	1.4776306 $= 4.534 \times (1 - 0.6741)$	scale: ± 0.016 (<i>absolute</i>)	LNupdg(t->Wb100%)">https://twiki.cern.ch/twiki/bin/view/LHCPhysics/SingleTopNNLORef#Single_top_quark_t_channel_crosstopratiotakenfromXSDB,BRtakenfromW->LNupdg(t->Wb100%)	—
TBbartoLplusNuBbar_s_channel_4FS	σ [pb]	2.3608196 $= 7.244 \times (1 - 0.6741)$	scale: ± 0.016 (<i>absolute</i>)	LNupdg(t->Wb100%)">https://twiki.cern.ch/twiki/bin/view/LHCPhysics/SingleTopNNLORef#Single_top_quark_t_channel_crosstopratiotakenfromXSDB,BRtakenfromW->LNupdg(t->Wb100%)	—
TbarBQ_t_channel_4FS	σ [pb]	87.2	scale: ± 0.1 (<i>absolute</i>) PDF + α_s: ± 0.2 (<i>absolute</i>) total: ± 0.3 (<i>absolute</i>) mass: $\pm - 0.1$ (<i>absolute</i>) beam energy: ± 0 (<i>absolute</i>) integration: ± 0.1 (<i>absolute</i>)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/SingleTopNNLORef	—

Continued on next page

Table 1 – continued					
Sample	Type	Value	Uncertainty	Reference	Comments
TBbarQ_t_channel_4FS	σ [pb]	145	scale: ± 0.6 (<i>absolute</i>) PDF + α_s : ± 0.8 (<i>absolute</i>) total: ± 0.9 (<i>absolute</i>) mass: ± 0.4 (<i>absolute</i>) beam energy: ± 0.1 (<i>absolute</i>) integration: ± 0.1 (<i>absolute</i>)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/SingleTopNNLORef	—
TbarBQtoLNu_t_channel_4FS	σ [pb]	23.34	total: ± 0.05875 (<i>absolute</i>)	XSDBTbarBQtoLNu-t-channel-4FS_TuneCP5_13p6TeV_powheg-madspin-pythia8modifiedOn2025-09-2209:07:43	—
TbarBQto2Q_t_channel_4FS	σ [pb]	46.73	—	XSDB	—
TBbarQtoLNu_t_channel_4FS	σ [pb]	38.6	—	XSDB	—
TBbarQto2Q_t_channel_4FS	σ [pb]	77.26	—	XSDB	—
TbarWplusto2L2Nu	σ [pb]	4.6726331 $= 87.9 \times (0.5005) \times (1 - 0.6741) \times (1 - 0.6741)$	scale: ± 0.1 (<i>absolute</i>) PDF + α_s : ± 2.4 (<i>absolute</i>) total: ± 0 (<i>absolute</i>) mass: ± 0 (<i>absolute</i>) beam energy: ± 0 (<i>absolute</i>)	Valuetakenfrom Wb100%">https://twiki.cern.ch/twiki/bin/view/LHCPhysics/SingleTopNNLORef,top/antitop%takenfromXSDB,BRstakenfrompdg,assumet->Wb100%	—
TbarWplusto4Q	σ [pb]	19.991326 $= 87.9 \times (0.5005) \times (0.6741) \times (0.6741)$	scale: ± 0.1 (<i>absolute</i>) PDF + α_s : ± 2.4 (<i>absolute</i>) total: ± 0 (<i>absolute</i>) mass: ± 0 (<i>absolute</i>) beam energy: ± 0 (<i>absolute</i>)	Valuetakenfrom Wb100%">https://twiki.cern.ch/twiki/bin/view/LHCPhysics/SingleTopNNLORef,top/antitop%takenfromXSDB,BRstakenfrompdg,assumet->Wb100%	—
TbarWplustoLNu2Q	σ [pb]	19.32999 $= 87.9 \times (0.5005) \times (((1 - 0.6741) \times (0.6741)) + ((0.6741) \times (1 - 0.6741)))$	scale: ± 0.1 (<i>absolute</i>) PDF + α_s : ± 2.4 (<i>absolute</i>) total: ± 0 (<i>absolute</i>) mass: ± 0 (<i>absolute</i>) beam energy: ± 0 (<i>absolute</i>)	Valuetakenfrom Wb100%">https://twiki.cern.ch/twiki/bin/view/LHCPhysics/SingleTopNNLORef,top/antitop%takenfromXSDB,BRstakenfrompdg,assumet->Wb100%	—
TWminusto2L2Nu	σ [pb]	4.6632971 $= 87.9 \times (0.4995) \times (1 - 0.6741) \times (1 - 0.6741)$	scale: ± 0.1 (<i>absolute</i>) PDF + α_s : ± 2.4 (<i>absolute</i>) total: ± 0 (<i>absolute</i>) mass: ± 0 (<i>absolute</i>) beam energy: ± 0 (<i>absolute</i>)	Valuetakenfrom Wb100%">https://twiki.cern.ch/twiki/bin/view/LHCPhysics/SingleTopNNLORef,top/antitop%takenfromXSDB,BRstakenfrompdg,assumet->Wb100%	—
TWminusto4Q	σ [pb]	19.951384 $= 87.9 \times (0.4995) \times (0.6741) \times (0.6741)$	scale: ± 0.1 (<i>absolute</i>) PDF + α_s : ± 2.4 (<i>absolute</i>) total: ± 0 (<i>absolute</i>) mass: ± 0 (<i>absolute</i>) beam energy: ± 0 (<i>absolute</i>)	Valuetakenfrom Wb100%">https://twiki.cern.ch/twiki/bin/view/LHCPhysics/SingleTopNNLORef,top/antitop%takenfromXSDB,BRstakenfrompdg,assumet->Wb100%	—
TWminustoLNu2Q	σ [pb]	19.291369 $= 87.9 \times (0.4995) \times (((1 - 0.6741) \times (0.6741)) + ((0.6741) \times (1 - 0.6741)))$	scale: ± 0.1 (<i>absolute</i>) PDF + α_s : ± 2.4 (<i>absolute</i>) total: ± 0 (<i>absolute</i>) mass: ± 0 (<i>absolute</i>) beam energy: ± 0 (<i>absolute</i>)	Valuetakenfrom Wb100%">https://twiki.cern.ch/twiki/bin/view/LHCPhysics/SingleTopNNLORef,top/antitop%takenfromXSDB,BRstakenfrompdg,assumet->Wb100%	—
TT	σ [pb]	923.6	scale: ± 10.8 (<i>absolute</i>) PDF + α_s : ± 22.8 (<i>absolute</i>) mass: ± 0.8 (<i>absolute</i>)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/TtbarNNLO#Updated_reference_cross_sections	—
TTto2L2Nu	σ [pb]	98.096304 $= 923.6 \times (1 - 0.6741) \times (1 - 0.6741)$	scale: ± 10.8 (<i>absolute</i>) PDF + α_s : ± 22.8 (<i>absolute</i>) mass: ± 0.8 (<i>absolute</i>)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/TtbarNNLO#Updated_reference_cross_sections	—
TTto4Q	σ [pb]	419.69382 $= 923.6 \times 0.6741 \times 0.6741$	scale: ± 10.8 (<i>absolute</i>) PDF + α_s : ± 22.8 (<i>absolute</i>) mass: ± 0.8 (<i>absolute</i>)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/TtbarNNLO#Updated_reference_cross_sections	—
TTtoLNu2Q	σ [pb]	405.80987 $= 923.6 \times 0.6741 \times (1 - 0.6741) \times 2$	scale: ± 10.8 (<i>absolute</i>) PDF + α_s : ± 22.8 (<i>absolute</i>) mass: ± 0.8 (<i>absolute</i>)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/TtbarNNLO#Updated_reference_cross_sections	—
TTZ_Zto2Q	σ [pb]	0.6603	total: ± 0.003767 (<i>absolute</i>)	XSDBTTZ-ZtoQQ-1Jets_TuneCP5_13p6TeV_amcatnlO5FXFX-pythia8modifiedOn2024-04-0319:49:29	—
TTWH	σ [pb]	0.001252	total: $\pm 4.375e - 07$ (<i>absolute</i>)	XSDBTTWH_TuneCP5_13p6TeV_madgraph-pythia8modifiedOn2024-04-0319:49:28	—
TTWW	σ [pb]	0.008203	total: $\pm 1.404e - 05$ (<i>absolute</i>)	XSDBTTWW_TuneCP5_13p6TeV_madgraph-madspin-pythia8modifiedOn2023-05-2621:19:17	—
TTWZ	σ [pb]	0.002715	total: $\pm 5.963e - 07$ (<i>absolute</i>)	XSDBTTWZ_TuneCP5_13p6TeV_madgraph-pythia8modifiedOn2025-09-1715:30:00	—
TTZH	σ [pb]	0.001288	total: $\pm 4.101e - 07$ (<i>absolute</i>)	XSDBTTZH_TuneCP5_13p6TeV_madgraph-pythia8modifiedOn2024-04-0319:49:34	—
TTZH_ZHto4B	σ [pb]	0.0001961	total: $\pm 4.663e - 08$ (<i>absolute</i>)	XSDBTTZH-ZHto4B_TuneCP5_13p6TeV_madgraph-pythia8modifiedOn2025-09-2209:07:41	—
TTZZ	σ [pb]	0.001579	total: $\pm 3.248e - 06$ (<i>absolute</i>)	XSDBTTZZ_TuneCP5_13p6TeV_madgraph-madspin-pythia8modifiedOn2023-05-2621:19:17	—
TTZZ_ZZto4B	σ [pb]	3.623e - 05	total: $\pm 1.217e - 08$ (<i>absolute</i>)	XSDBTTZZ-ZZto4B_TuneCP5_13p6TeV_madgraph-pythia8modifiedOn2025-09-2209:07:41	—

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Table 1 – continued					
Sample	Type	Value	Uncertainty	Reference	Comments
TTHto2B_M125	σ [pb]	0.331968 = $0.57 \times (0.5824)$	QCD scale: $+6\%$ -9.3% (relative) PDF + α_s : $\pm 3.5\%$ (relative) pdf: $\pm 3\%$ (relative) α_s : ± 2 (absolute)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LHCHWG136TeVxsec_extrap	—
TTHtoNon2B_M125	σ [pb]	0.238032 = $0.57 \times (1 - 0.5824)$	QCD scale: $+6\%$ -9.3% (relative) PDF + α_s : $\pm 3.5\%$ (relative) pdf: $\pm 3\%$ (relative) α_s : ± 2 (absolute)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LHCHWG136TeVxsec_extrap	—
TTH_Hto2Mu	σ [pb]	0.000124032 = $0.57 \times (0.0002176)$	QCD scale: $+6\%$ -9.3% (relative) PDF + α_s : $\pm 3.5\%$ (relative) pdf: $\pm 3\%$ (relative) α_s : ± 2 (absolute) theory (BR): $+1.23\%$ -1.2% (relative) m_q (BR): $+0.97\%$ -0.99% (relative) α_s (BR): $+0.59\%$ -0.64% (relative)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LHCHWG136TeVxsec_extrap	—
VBFHto2Mu_M120	σ [pb]	0.0009880994 = $4.078 \times (0.0002423)$	QCD scale: $+0.5\%$ -0.3% (relative) PDF + α_s : $\pm 2.1\%$ (relative) pdf: $\pm 2.1\%$ (relative) α_s : ± 0.5 (absolute) theory (BR): $+1.23\%$ 1.23% (relative) m_q (BR): $+1.07\%$ -1.09% (relative) α_s (BR): $+0.69\%$ -0.70% (relative)	https://arxiv.org/pdf/2402.09955forXSandhttps://twiki.cern.ch/twiki/bin/view/LHCPhysics/CERNYellowReportPageBRforBR	—
VBFHto2Mu_M125	σ [pb]	0.0008873728 = $4.078 \times (0.0002176)$	QCD scale: $+0.5\%$ -0.3% (relative) PDF + α_s : $\pm 2.1\%$ (relative) pdf: $\pm 2.1\%$ (relative) α_s : ± 0.5 (absolute) theory (BR): $+1.23\%$ -1.2% (relative) m_q (BR): $+0.97\%$ -0.99% (relative) α_s (BR): $+0.59\%$ -0.64% (relative)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LHCHWG136TeVxsec_extrapandhttps://twiki.cern.ch/twiki/bin/view/LHCPhysics/CERNYellowReportPageBRforBR	—
VBFHto2Mu_M130	σ [pb]	0.0007654406 = $4.078 \times (0.0001877)$	QCD scale: $+0.5\%$ -0.3% (relative) PDF + α_s : $\pm 2.1\%$ (relative) pdf: $\pm 2.1\%$ (relative) α_s : ± 0.5 (absolute) theory (BR): $+1.22\%$ 1.21% (relative) m_q (BR): $+0.86\%$ -0.82% (relative) α_s (BR): $+0.52\%$ -0.47% (relative)	https://arxiv.org/pdf/2402.09955forXSandhttps://twiki.cern.ch/twiki/bin/view/LHCPhysics/CERNYellowReportPageBRforBR	—
VBFHto2Mu	σ [pb]	0.0008873728 = $4.078 \times (0.0002176)$	QCD scale: $+0.5\%$ -0.3% (relative) PDF + α_s : $\pm 2.1\%$ (relative) pdf: $\pm 2.1\%$ (relative) α_s : ± 0.5 (absolute) theory (BR): $+1.23\%$ -1.2% (relative) m_q (BR): $+0.97\%$ -0.99% (relative) α_s (BR): $+0.59\%$ -0.64% (relative)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LHCHWG136TeVxsec_extrapandhttps://twiki.cern.ch/twiki/bin/view/LHCPhysics/CERNYellowReportPageBRforBR	—
VBFHtoTauTau_M125	σ [pb]	0.25577216 = $4.078 \times (0.06272)$	QCD scale: $+0.5\%$ -0.3% (relative) PDF + α_s : $\pm 2.1\%$ (relative) pdf: $\pm 2.1\%$ (relative) α_s : ± 0.5 (absolute)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LHCHWG136TeVxsec_extrap	—
VBFHtoWWTo2L2Nu_M125	σ [pb]	0.28392447 = $4.078 \times (0.2137) \times (3 \times 0.1086)$	QCD scale: $+0.5\%$ -0.3% (relative) PDF + α_s : $\pm 2.1\%$ (relative) pdf: $\pm 2.1\%$ (relative) α_s : ± 0.5 (absolute)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LHCHWG136TeVxsec_extrap	—
WminusH_Hto2Mu_WtoAll	σ [pb]	0.00012353152 = $0.5677 \times (0.0002176)$	QCD scale: $+0.4\%$ -0.7% (relative) PDF + α_s : $\pm 1.8\%$ (relative) pdf: ± 1.6 (absolute) α_s : ± 0.9 (absolute) theory (BR): $+1.23\%$ -1.2% (relative) m_q (BR): $+0.97\%$ -0.99% (relative) α_s (BR): $+0.59\%$ -0.64% (relative)	https://indico.cern.ch/event/1119741/contributions/4715908/attachments/2383849/4073592/YR4_13p6_VH_update.pdf	—

Table 1 – continued					
Sample	Type	Value	Uncertainty	Reference	Comments
WplusH_Hto2Mu_WtoA1l	σ [pb]	0.00019342464 = 0.8889 $\times$ (0.0002176)	QCD scale: +0.4% −0.7% <i>(relative)</i> PDF + α_s: $\pm 1.8\%$ <i>(relative)</i> pdf: ± 1.6 <i>(absolute)</i> α_s : ± 0.9 <i>(absolute)</i> theory (BR): +1.23% −1.2% <i>(relative)</i> m_q (BR): +0.97% −0.99% <i>(relative)</i> α_s (BR): +0.59% −0.64% <i>(relative)</i>	https://indico.cern.ch/event/1119741/contributions/4715908/attachments/2383849/4073592/YR4_13p6_VH_update.pdf	—
WtoLNu_OJ_amcatnloFXFX	σ [pb]	55760	total: ± 75.39 <i>(absolute)</i>	XSDBWtoLNu-2Jets_OJ_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0320:09:44	—
WtoLNu_1J_amcatnloFXFX	σ [pb]	9529	total: ± 56.7 <i>(absolute)</i>	XSDBWtoLNu-2Jets_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0320:09:46	—
WtoLNu_1J_madgraphMLM	σ [pb]	9084	total: ± 24.69 <i>(absolute)</i>	XSDBWtoLNu-4Jets_1J_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0320:10:09	—
WtoLNu_2J_amcatnloFXFX	σ [pb]	3532	total: ± 34.81 <i>(absolute)</i>	XSDBWtoLNu-2Jets_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0320:09:47	—
WtoLNu_2J_madgraphMLM	σ [pb]	2925	total: ± 8.555 <i>(absolute)</i>	XSDBWtoLNu-4Jets_2J_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0320:10:12	—
WtoLNu_3J_madgraphMLM	σ [pb]	861.7	total: ± 2.624 <i>(absolute)</i>	XSDBWtoLNu-4Jets_3J_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0320:10:13	—
WtoLNu_4J_madgraphMLM	σ [pb]	416.5	total: ± 1.28 <i>(absolute)</i>	XSDBWtoLNu-4Jets_4J_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0320:10:17	—
WtoLNu_NNLO_QCD_NLO_EW_one_flav	σ [pb]	21141.7	electron neutrino: +1.2% -1.1% +−0.8% <i>(relative)</i> positron neutrino: +1.1% -1.4% +−0.7% <i>(relative)</i>	https://twiki.cern.ch/twiki/bin/viewauth/CMS/MATRIXCrossSectionsat13p6TeV	—
WtoLNu_NNLO_QCD+NLO_EW	σ [pb]	63425.1	electron neutrino: +1.2% -1.1% +−0.8% <i>(relative)</i> positron neutrino: +1.1% -1.4% +−0.7% <i>(relative)</i>	https://twiki.cern.ch/twiki/bin/viewauth/CMS/MATRIXCrossSectionsat13p6TeV	—
WtoLNu_MLNu_0to120_HT_40To100_madgraph	σ [pb]	4254	total: ± 12.08 <i>(absolute)</i>	XSDBWtoLNu-4Jets_MLNu-0to120_HT-40to100_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0320:10:25	—
WtoLNu_MLNu_0to120_HT_100To400_madgraph	σ [pb]	1626	total: ± 4.823 <i>(absolute)</i>	XSDBWtoLNu-4Jets_MLNu-0to120_HT-100to400_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0320:10:18	—
WtoLNu_MLNu_0to120_HT_400To800_madgraph	σ [pb]	59.99	total: ± 0.1803 <i>(absolute)</i>	XSDBWtoLNu-4Jets_MLNu-0to120_HT-400to800_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0320:10:23	—
WtoLNu_MLNu_0to120_HT_800To1500_madgraph	σ [pb]	6.23	total: ± 0.01876 <i>(absolute)</i>	XSDBWtoLNu-4Jets_MLNu-0to120_HT-800to1500_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0320:10:28	—
WtoLNu_MLNu_0to120_HT_1500To2500_madgraph	σ [pb]	0.4477	total: ± 0.00135 <i>(absolute)</i>	XSDBWtoLNu-4Jets_MLNu-0to120_HT-1500to2500_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0320:10:20	—
WtoLNu_PTLNu_40to100_1J_amcatnloFXFX	σ [pb]	4427	total: ± 28.3 <i>(absolute)</i>	XSDBWtoLNu-2Jets_PTLNu-40to100_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2025-09-19	—
WtoLNu_PTLNu_40to100_2J_amcatnloFXFX	σ [pb]	1598	total: ± 16.36 <i>(absolute)</i>	XSDBWtoLNu-2Jets_PTLNu-40to100_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2025-09-19	—
WtoLNu_PTLNu_100to200_1J_amcatnloFXFX	σ [pb]	368.2	total: ± 1.926 <i>(absolute)</i>	XSDBWtoLNu-2Jets_PTLNu-100to200_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2025-09-19	—
WtoLNu_PTLNu_100to200_2J_amcatnloFXFX	σ [pb]	421.9	total: ± 3.533 <i>(absolute)</i>	XSDBWtoLNu-2Jets_PTLNu-100to200_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2025-09-19	—
WtoLNu_PTLNu_200to400_1J_amcatnloFXFX	σ [pb]	25.6	total: ± 0.1207 <i>(absolute)</i>	XSDBWtoLNu-2Jets_PTLNu-200to400_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2025-09-19	—
WtoLNu_PTLNu_200to400_2J_amcatnloFXFX	σ [pb]	54.77	total: ± 0.3764 <i>(absolute)</i>	XSDBWtoLNu-2Jets_PTLNu-200to400_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2025-09-19	—
WtoLNu_PTLNu_400to600_1J_amcatnloFXFX	σ [pb]	0.8785	total: ± 0.003223 <i>(absolute)</i>	XSDBWtoLNu-2Jets_PTLNu-400to600_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2025-09-19	—
WtoLNu_PTLNu_400to600_2J_amcatnloFXFX	σ [pb]	3.119	total: ± 0.01749 <i>(absolute)</i>	XSDBWtoLNu-2Jets_PTLNu-400to600_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2025-09-19	—
WtoLNu_PTLNu_600_1J_amcatnloFXFX	σ [pb]	0.1053	total: ± 0.0003437 <i>(absolute)</i>	XSDBWtoLNu-2Jets_PTLNu-600_1J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2025-09-19	—
WtoLNu_PTLNu_600_2J_amcatnloFXFX	σ [pb]	0.5261	total: ± 0.002417 <i>(absolute)</i>	XSDBWtoLNu-2Jets_PTLNu-600_2J_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2025-09-19	—
WToMuNu_M_200	σ [pb]	7.693	total: ± 0.0408 <i>(absolute)</i>	XSDBWtoMuNu_M-200_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2614:05:35	—
WToTauNu_M_200	σ [pb]	7.753	total: ± 0.04088 <i>(absolute)</i>	XSDBWtoNuTau_M-200_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2614:05:40	—
WW	σ [pb]	80.23	total: ± 0.3733 <i>(absolute)</i>	XSDBWW_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2614:02:32	—
WWto2L2Nu	σ [pb]	11.79	total: ± 0.004216 <i>(absolute)</i>	XSDBWWto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0320:06:05	—
WWW	σ [pb]	0.2328	total: ± 0.0001247 <i>(absolute)</i>	XSDBWWW_4F_TuneCP5_13p6TeV_amcatnlo-madspin-pythia8modifiedOn2024-03-0204:47:30	—

Table 1 – continued					
Sample	Type	Value	Uncertainty	Reference	Comments
WWZ	σ [pb]	0.1851	total: $\pm 9.482e - 05$ (<i>absolute</i>)	XSDBWZ_4F_TuneCP5_13p6TeV_ amcatnlo-pythia8modifiedOn2023-05-2621:19:27	—
WZZ	σ [pb]	0.06206	total: $\pm 3.689e - 05$ (<i>absolute</i>)	XSDBWZZ_4F_TuneCP5_13p6TeV_ amcatnlo-pythia8modifiedOn2023-05-2621:19:28	—
WZ	σ [pb]	29.1	total: ± 0.1318 (<i>absolute</i>)	XSDBWZ_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2614:02:34	—
WZto2L2Q_powheg	σ [pb]	7.568	total: ± 0.003908 (<i>absolute</i>)	XSDBWZto2L2Q_TuneCP5_13p6TeV_ powheg-pythia8modifiedOn2024-04-0320:06:15	—
WZto3LNu_amcatnloFXFX	σ [pb]	5.315	total: ± 0.01577 (<i>absolute</i>)	XSDBWZto3LNu-1Jets-4FS_TuneCP5_13p6TeV_ amcatnloFXFX-pythia8modifiedOn2024-04-0320:06:16	—
WZtoL3Nu_powheg	σ [pb]	4.924	total: ± 0.00237 (<i>absolute</i>)	XSDBWZto3LNu_TuneCP5_13p6TeV_ powheg-pythia8modifiedOn2024-04-0320:06:18	—
WZto3LNu_powheg	σ [pb]	4.924	total: ± 0.00237 (<i>absolute</i>)	XSDBWZto3LNu_TuneCP5_13p6TeV_ powheg-pythia8modifiedOn2025-09-19	—
WZtoL3Nu_amcatnloFXFX	σ [pb]	3.445	total: ± 0.01204 (<i>absolute</i>)	XSDBWZtoL3Nu-1Jets-4FS_TuneCP5_13p6TeV_ amcatnloFXFX-pythia8modifiedOn2024-04-0320:06:23	—
WZtoLNu2Q_amcatnloFXFX	σ [pb]	12.11	total: ± 0.04269 (<i>absolute</i>)	XSDBWZtoLNu2Q-1Jets-4FS_TuneCP5_13p6TeV_ amcatnloFXFX-pythia8modifiedOn2024-04-0320:06:25	—
WZtoLNu2Q_powheg	σ [pb]	15.87	total: ± 0.007874 (<i>absolute</i>)	XSDBWZtoLNu2Q_TuneCP5_13p6TeV_ powheg-pythia8modifiedOn2024-04-0320:06:26	—
WWtoLNu2Q	σ [pb]	48.94	total: ± 0.0175 (<i>absolute</i>)	XSDBWWtoLNu2Q_TuneCP5_13p6TeVmodifiedOn2024-04-0320:06:13	—
WWto4Q-1Jets	σ [pb]	54.82	total: ± 0.1846 (<i>absolute</i>)	XSDBWWto4Q-1Jets-4FS_TuneCP5_13p6TeVmodifiedOn2024-04-0320:06:06	—
WWto4Q	σ [pb]	50.79	total: ± 0.01816 (<i>absolute</i>)	XSDBWWto4Q_TuneCP5_13p6TeVmodifiedOn2024-04-0320:06:10	—
WWtoLNu2Q-1Jets	σ [pb]	52.98	total: ± 0.1795 (<i>absolute</i>)	XSDBWWtoLNu2Q-1Jets-4FS_TuneCP5_13p6TeVmodifiedOn2024-04-0320:06:12	—
ZH_Hbb_Z11	σ [pb]	0.05550817 = $9.439e - 1 \times 0.5824 \times 3 \times 0.033658$	QCD scale: ± 0.5 (<i>absolute</i>) PDF + α_s: $\pm - 1.6$ (<i>absolute</i>) pdf: $\pm - 1.3$ (<i>absolute</i>) α_s : $\pm - 0.9$ (<i>absolute</i>)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LHCHWG136TeVxsec_extrap	—
ZH_Hbb_Zqq	σ [pb]	0.38431989 = $9.439e - 1 \times 0.5824 \times 0.69911$	QCD scale: ± 0.5 (<i>absolute</i>) PDF + α_s: $\pm - 1.6$ (<i>absolute</i>) pdf: $\pm - 1.3$ (<i>absolute</i>) α_s : $\pm - 0.9$ (<i>absolute</i>)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LHCHWG136TeVxsec_extrap	—
ZH_Hto2Mu_ZtoA11	σ [pb]	0.00020539264 = $0.9439 \times (0.0002176)$	QCD scale: ± 0.5 (<i>absolute</i>) PDF + α_s: $\pm - 1.6$ (<i>absolute</i>) pdf: $\pm - 1.3$ (<i>absolute</i>) α_s : $\pm - 0.9$ (<i>absolute</i>) theory (BR): $+1.23\%$ (<i>relative</i>) -1.2% m_q (BR): $+0.97\%$ (<i>relative</i>) -0.99% α_s (BR): $+0.59\%$ (<i>relative</i>) -0.64%	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LHCHWG136TeVxsec_extrap	—
ZHto2Tau	σ [pb]	0.059201408 = 0.9439×0.06272	QCD scale: ± 0.5 (<i>absolute</i>) PDF + α_s: $\pm - 1.6$ (<i>absolute</i>) pdf: $\pm - 1.3$ (<i>absolute</i>) α_s : $\pm - 0.9$ (<i>absolute</i>) theory (BR): $+1.17\%$ (<i>relative</i>) -1.16% m_q (BR): $+0.98\%$ (<i>relative</i>) -0.99% α_s (BR): $+0.62\%$ (<i>relative</i>) -0.62%	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LHCHWG136TeVxsec_extrapandBRfromhttps://twiki.cern.ch/twiki/bin/view/LHCPhysics/CERNYellowReportPageBR	—
ZHto2Tau_Filtered	σ [pb]	0.023277994 = $0.9439 \times 0.06272 \times 0.3932$	QCD scale: ± 0.5 (<i>absolute</i>) PDF + α_s: $\pm - 1.6$ (<i>absolute</i>) pdf: $\pm - 1.3$ (<i>absolute</i>) α_s : $\pm - 0.9$ (<i>absolute</i>) theory (BR): $+1.17\%$ (<i>relative</i>) -1.16% m_q (BR): $+0.98\%$ (<i>relative</i>) -0.99% α_s (BR): $+0.62\%$ (<i>relative</i>) -0.62% filter efficiency (BR): $\pm 0.8\%$ (<i>relative</i>)	https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LHCHWG136TeVxsec_extrapandBRfromhttps://twiki.cern.ch/twiki/bin/view/LHCPhysics/CERNYellowReportPageBR*genXsecanalyzerfiltereff	—
Zto2Nu_HT_100To200	σ [pb]	273.7	total: ± 0.8085 (<i>absolute</i>)	XSDBZto2Nu-4Jets_HT-100to200_TuneCP5_13p6TeV_ madgraphMLM-pythia8modifiedOn2024-04-0320:13:00	—
Zto2Nu_HT_200To400	σ [pb]	75.96	total: ± 0.2275 (<i>absolute</i>)	XSDBZto2Nu-4Jets_HT-200to400_TuneCP5_13p6TeV_ madgraphMLM-pythia8modifiedOn2024-04-0320:13:04	—
Zto2Nu_HT_400To800	σ [pb]	13.19	total: ± 0.03963 (<i>absolute</i>)	XSDBZto2Nu-4Jets_HT-400to800_TuneCP5_13p6TeV_ madgraphMLM-pythia8modifiedOn2024-04-0320:13:07	—
Zto2Nu_HT_800To1500	σ [pb]	1.364	total: ± 0.004118 (<i>absolute</i>)	XSDBZto2Nu-4Jets_HT-800to1500_TuneCP5_13p6TeV_ madgraphMLM-pythia8modifiedOn2024-04-0320:13:09	—

Table 1 – continued

Sample	Type	Value	Uncertainty	Reference	Comments
Zto2Nu_HT_1500To2500	σ [pb]	0.09865	total: ± 0.0002974 (<i>absolute</i>)	XSDBZto2Nu-4Jets_HT-1500to2500_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0320:13:02	—
Zto2Nu_HT_2500ToInf	σ [pb]	0.006699	total: $\pm 2.019e - 05$ (<i>absolute</i>)	XSDBZto2Nu-4Jets_HT-2500_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2024-04-0320:13:06	—
Zto2Q_HT_100to400	σ [pb]	6328	total: ± 17.95 (<i>absolute</i>)	XSDBZto2Q-4Jets_Bin-HT-100to400_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2025-09-2209:08:33	—
Zto2Q_HT_400to800	σ [pb]	145.1	total: ± 0.4362 (<i>absolute</i>)	XSDBZto2Q-4Jets_Bin-HT-400to800_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2025-09-2209:08:33	—
Zto2Q_HT_800to1500	σ [pb]	12.9	total: ± 0.01236 (<i>absolute</i>)	XSDBZto2Q-4Jets_Bin-HT-800to1500_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2026-01-2520:47:07	—
Zto2Q_HT_1500to2500	σ [pb]	0.8541	total: $\pm 5.449e - 05$ (<i>absolute</i>)	XSDBZto2Q-4Jets_Bin-HT-1500to2500_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2026-01-2520:47:08	—
Zto2Q_HT_2500toInf	σ [pb]	0.05684	total: ± 0.0008198 (<i>absolute</i>)	XSDBZto2Q-4Jets_Bin-HT-2500_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2026-01-2520:47:08	—
Zto2Q_HT_200to400	σ [pb]	1082	total: ± 9.886 (<i>absolute</i>)	XSDBZto2Q-4Jets_HT-200to400_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2023-05-2621:19:45	—
Zto2Q_HT_400to600	σ [pb]	124.1	total: ± 1.163 (<i>absolute</i>)	XSDBZto2Q-4Jets_HT-400to600_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2023-05-2621:19:45	—
Zto2Q_HT_600to800	σ [pb]	27.28	total: ± 0.2549 (<i>absolute</i>)	XSDBZto2Q-4Jets_HT-600to800_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2023-05-2621:19:46	—
Zto2Q_HT_800toInf	σ [pb]	14.57	total: ± 0.1378 (<i>absolute</i>)	XSDBZto2Q-4Jets_HT-800_TuneCP5_13p6TeV_madgraphMLM-pythia8modifiedOn2023-05-2621:19:46	—
ZZ	σ [pb]	12.75	total: ± 0.0649 (<i>absolute</i>)	XSDBZZ_TuneCP5_13p6TeV_pythia8modifiedOn2023-03-2614:05:37	—
ZZto2L2Nu_powheg	σ [pb]	1.031	total: ± 0.0005268 (<i>absolute</i>)	XSDBZZto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0320:06:52	—
ZZto2L2Q_amcatnloFXFX	σ [pb]	3.86	total: ± 0.01259 (<i>absolute</i>)	XSDBZZto2L2Q-1Jets_TuneCP5_13p6TeV_amcatnloFXFX-pythia8modifiedOn2024-04-0320:12:33	—
ZZto2L2Q_powheg	σ [pb]	6.788	total: ± 0.003501 (<i>absolute</i>)	XSDBZZto2L2Q_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0320:08:04	—
ZZto2Nu2Q_powheg	σ [pb]	4.826	total: ± 0.001296 (<i>absolute</i>)	XSDBZZto2Nu2Q_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0320:08:39	—
ZZto4L_powheg	σ [pb]	1.39	total: ± 0.0007001 (<i>absolute</i>)	XSDBZZto4L_TuneCP5_13p6TeV_powheg-pythia8modifiedOn2024-04-0320:11:16	—
ZZZ	σ [pb]	0.01591	total: $\pm 7.828e - 06$ (<i>absolute</i>)	XSDBZZZ_TuneCP5_13p6TeV_amcatnlo-pythia8modifiedOn2023-05-2621:19:33	—
DYto2L_2Jets_EWK_MLL_50_L0	σ [pb]	13.07	total: ± 0.003645 (<i>absolute</i>)	XSDBDYto2L-2Jets-EWK_MLL-50_TuneCP5_13p6TeV_madgraph-pythia8modifiedOn2026-01-2600:01:51	—
EWK_2L2J_MLL50_MJJ120	σ [pb]	1.418	total: ± 0.002142 (<i>absolute</i>)	fromgridpackgenerationlogtakenfro/cvmfs/cms.cern.ch/phys_generator/gridpacks/RunIII/13p6TeV/slc7_amd64_gcc10/MadGraph5_aMCatNLO/LLJJ_50-Inf_EWK_SM_5f_L0_slc7_amd64_gcc10_CMSSW_12_4_8_tarball.tar.xzandwiththissetting	—
EWK_2Mu2J_MLL105to160_MJJ120	σ [pb]	0.0635	total: ± 0.0003071 (<i>absolute</i>)	fromgridpackgenerationlogtakenfrom/cvmfs/cms.cern.ch/phys_generator/gridpacks/RunIII/13p6TeV/slc7_amd64_gcc10/MadGraph5_aMCatNLO/LLJJ_105-160_EWK_SM_5f_L0_slc7_amd64_gcc10_CMSSW_12_4_8_tarball.tar.xzandwiththissetting	—
TTLNu_EWK	σ [pb]	0.01769	total: $\pm 5.69e - 06$ (<i>absolute</i>)	XSDBTTLNu-EWK_TuneCP5_13p6TeV_amcatnlo-pythia8modifiedOn2024-12-2008:23:11	—
WZto3LNu_2Jets_EWK_QCD	σ [pb]	0.5437	total: ± 0.0001215 (<i>absolute</i>)	XSDBZto3LNu-2Jets_EWK-QCD_TuneCP5_13p6TeV_madgraph-madspin-pythia8modifiedOn2024-12-2008:28:09	—
WmWmJJ_EWK	σ [pb]	0.000122	total: $\pm 2.832e - 05$ (<i>absolute</i>)	XSDBWmWmJJ-EWK_TuneCP5_13p6TeV-powheg-pythia8modifiedOn2025-09-2209:07:36	—